Heuristic Usability Evaluation
OOPP Group 1

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1 INTRODUCTION

This report documents an evaluation of the user interface design of our OOPP team's project, with the aim of identifying and resolving any usability issues. Following the methods outlined in [3], we evaluated a prototype of our task management application, Talio. In this report, we will provide an analysis of the application's evaluated state, its problems, and the redesigned UI. To familiarize the reader with our prototype, we will now illustrate the different windows that a user may have encountered during testing.

The application starts-up with the screen in figure 1, providing two options: clicking the "User log in" button leads to the screen in figure 2, while clicking the "Admin log in" button leads to the one in figure 3. Both of them allow connecting to a desired server and returning to the previous screen, but the admin is additionally required a server password.

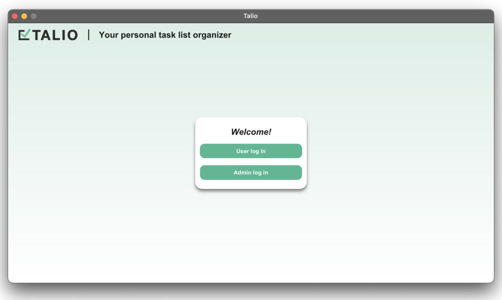

Figure 1: Start-up screen

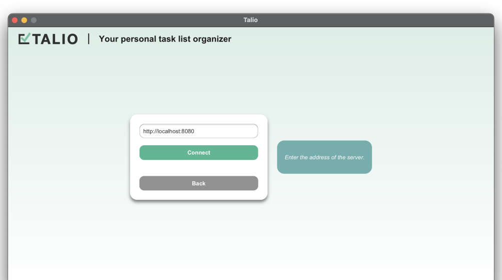

Figure 2: User log in

After connecting to the server, the workspace corresponding to the chosen role appears. An example user workspace is shown in figure 4. Both user and admin variants look and behave similarly,

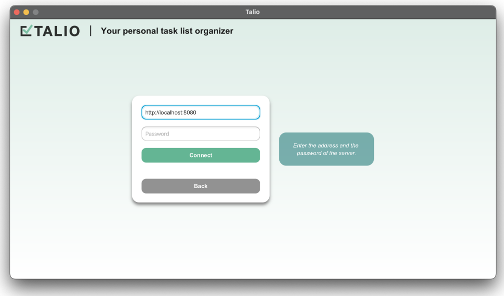

Figure 3: Admin log in

with two differences: 1) upon initialization, the user is shown no boards in the overview (since they haven't yet created any), while the admin is provided with all existing server boards; and 2) in the left-side board list, the user sees "door" icons to leave boards, while the admin sees "trash bin" icons to delete server boards.

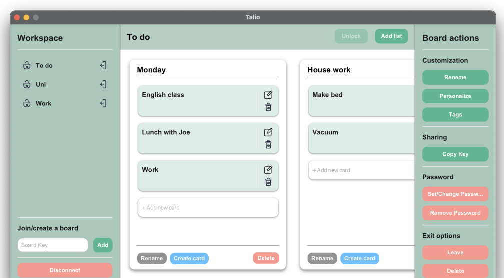

Figure 4: Workspace

The workspace presents a number of features, via the buttons in the right-side of the screen: adding a new list, renaming the board, personalizing it, creating tags which can be added to cards, copying the key of the board, setting/ changing/removing the password of the board, and finally leaving or deleting the board.

Clicking the "Add list" button adds a new list titled "New list" to the board directly. Clicking the "Rename" button on any list makes the title editable, allowing name changes by pressing enter to confirm (figure 5).

Clicking the "Delete" button of any list opens a pop-up similar to the first one in figure 6, to confirm the deletion. The white "Add new card" button below the final card in any list can be used to quickly add a simple task by just providing a title. Alternatively, the blue "Create card" button opens a pop-up for adding a more complex card (figure 7).

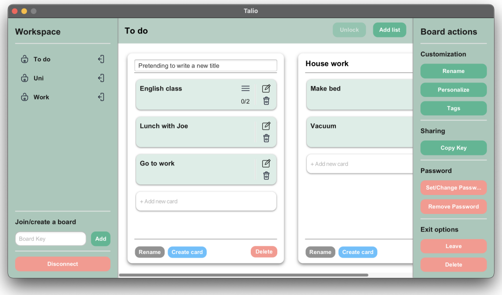

Figure 5: Renaming lists

Each card may also display indications that it has, for instance, a description or nested sub-tasks directly in the overview. The "pencil" and "trash bin" icons in the right-side of any card can be used to edit the card (which opens a window similar to the one in figure 7), or delete it, respectively.

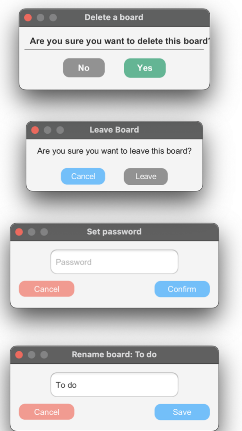

Figure 6: Pop-ups with different purposes

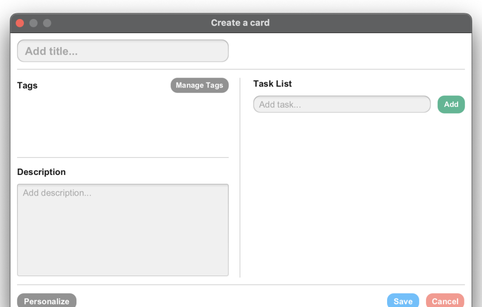

Figure 7: Create a card

Double-clicking a card pops-up a window similar to the one in figure 8, allowing you to view its details.

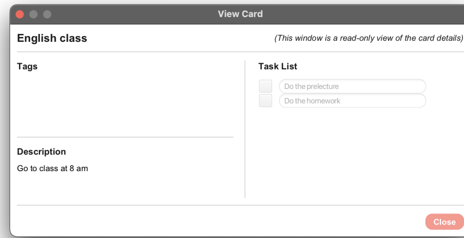

Figure 8: View a card

The "Tags" button pops-up the "Tag Overview" window, seen at the top of figure 9, which shows a list of all existing board tags, allowing you to edit, delete or add tags. The "Add a new tag" button redirects to the "Add Tag" window, seen at the bottom of the same figure.

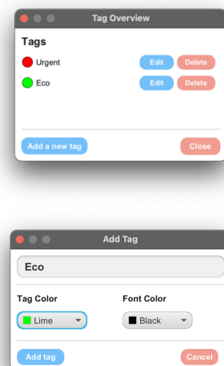

Figure 9: Tag windows

The "Personalize" button opens a customization pop-up (figure 10), which allows you to change the colors of boards and lists, reset them, and also add color presets for cards.

2 METHODS

This section describes the methodology used when executing the evaluation.

2.1 Experts

We recruited 4 experts from OOPP Group 29. As Computer Science and Engineering students who have used other scheduling apps extensively, they are experienced with software like Talio as both users and professionals, therefore being the perfect choice for our evaluation.

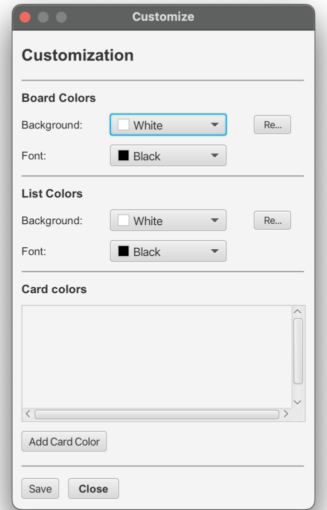

Figure 10: Customization window

2.2 Procedure

We held the procedure in the Drebbelweg building, an open environment at TU Delft, thus proper for our evaluation, which consisted of two parts.

2.2.1 First stage. During this phase of the evaluation, every expert was guided by a member of our team, who provided them with a Google Form with 7 tasks focused on basic capabilities of our prototype. These were read to each evaluator by their assistant, to make sure that everything is clear:

(1) Connect to the default server as a user.
(2) Create 3 boards.
(3) Rename the last board you created.
(4) Add 3 lists to this board.
(5) Rename the last list you created.
(6) Add 3 cards to this list.
(7) Drag and drop the cards you created between lists on the board.

Each of the 7 questions accepted an answer from a scale of 1 to 5 about how difficult that task is (1 was marked in the form as "clear and intuitive", while 5 as "difficult and annoying"). Each evaluator had to answer the form individually to minimize biased results, and in the case of an answer of at least 3 (i.e., poorly implemented feature), they were also prompted to motivate their choice.

2.2.2 Second stage. After going through the typical usage scenario in stage 1, the experts were now asked to test the app freely, for a period of 1 hour, and to describe the problems they found in another Google Form, containing clear instructions and rules devised for replicability purposes. They were required to report the problems in the following format, which we took from [1]:

(1) Problem description: a brief description of the problem;
(2) Likely/actual difficulties: the anticipated difficulties that the user will encounter as a consequence of this problem;
(3) Specific contexts: the specific context in which the problem may occur;
(4) Assumed causes: description of the cause(s) of the problem.

Lastly, we required the evaluators to write the "heuristic" that fits each problem, following the 10 heuristics described in [2]. For this, we made sure that the experts were familiar with the heuristics beforehand, and we included them in the form for quick reference. Due to word limitations, we will only mention their titles, but they are all detailed in [2]:

(1) Visibility of system status;
(2) Match between system and the real world;
(3) User control and freedom;
(4) Consistency and standards;
(5) Error prevention;
(6) Recognition rather than recall;
(7) Flexibility and efficiency of use;
(8) Aesthetic and minimalist design;
(9) Help users recognize, diagnose, and recover from errors;
(10) Help and documentation.

In situations where the experts struggled with the interface, we gave them hints of how to achieve what they want, and we also noted down any comments they had, without responding.

2.3 Measures (Data collection)

Raw results were collected via Google Forms automatically. Additionally, comments and remarks from the experts were noted on paper.

3 RESULTS

In this section, we will list and analyze the identified problems, based on the two evaluation stages, while also prioritizing them accordingly.

3.1 Stage 1 problems

We processed the results of this stage by taking into consideration the form questions with responses of difficulty at least 3. These are shown in figures 11 and 12, which helped us extract the following problems:

- **Problem 1:** Adding lists with the predefined "New list" title is inconvenient as it requires an extra step to rename it to your desired title (heuristic: "Flexibility and efficiency of use");
- **Problem 2:** It is not explicit that users have to press Enter to save the new title when renaming a list, and moving the cursor doesn't allow pressing Enter to save anymore (heuristic: "Recognition rather than recall").

3.2 Stage 2 problems

Here the collected problems were processed as follows: we took those functionality-related out of consideration (as they were out of scope for our evaluation), removed duplicates and split mistakenly grouped problems.

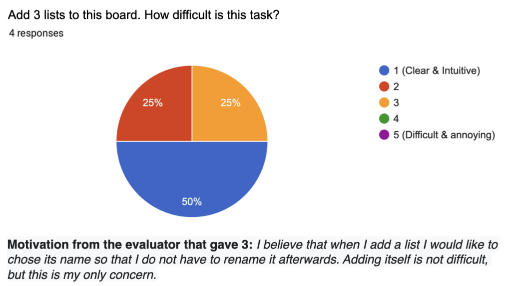

Figure 11: Results of question (4)

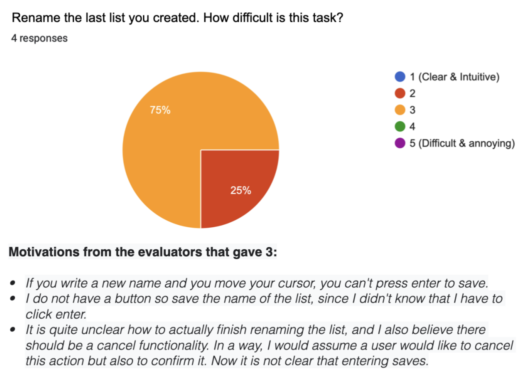

Figure 12: Results of question (5)

Problem 3

Identified Heuristic: Flexibility and efficiency of use

(1) Problem description: The board key is the same as the title;
(2) Likely/actual difficulties: The user may be confused as to whether the title is also the key of the board or they are two distinct entities;
(3) Specific contexts: If you rename the board to something completely unrelated, when sharing the key, it might cause confusion (a board having "Company" as the invite key can be renamed to "Personal");
(4) Assumed causes: Not enough consideration for people who are unfamiliar with scheduling apps.

Problem 4

Identified Heuristics: Help and documentation

(1) Problem description: Users might be confused by the "Task List" section inside of a task;
(2) Likely/actual difficulties: The card is already added to a list, so the user can get confused as to which Task List the label is referring to;
(3) Specific contexts: Whenever the user views or edits a card;
(4) Assumed causes: Oversight during design.

Problem 5

Identified Heuristic: Help and documentation

(1) Problem description: There is no help screen to help me understand the app;
(2) Likely/actual difficulties: I may have to go through multiple screens if I want to achieve a frequent task, like adding a tag to a card;
(3) Specific contexts: When I want to add a tag, I first have to open the card editing pop-up by mouse, then the manage tags pop-up, and then I can pick a tag. While this is fine by mouse, it could be quicker with a shortcut.
(4) Assumed causes: Not providing a help screen of frequently-used shortcuts.

Problem 6

Identified Heuristic: Consistency and standards

(1) Problem description: Many pop-up screens have different designs.
(2) Likely/actual difficulties: It can be difficult to adapt to the way different screens look at first glance.
(3) Specific contexts: The customization window lacks the same colors that other windows have, and when deleting or leaving a board you get different pop-up styles.
(4) Assumed causes: Not taking consistency into consideration.

3.3 Other problems

Based on the comments of the evaluators and the tendencies we noticed in them while using the UI, we further found these problems:

- **Problem 7:** Unfamiliar users may spend time identifying buttons in the right side of the workspace (heuristic: "Recognition rather than recall");
- **Problem 8:** The positioning of buttons on the right side of pop-ups is inflexible, as we noticed that most of the evaluators had a tendency of clicking to the left (heuristic: "Flexibility and efficiency of use").

3.4 Prioritization

After a team discussion, we established the severity of a problem based on 2 criteria: the frequency and the impact. Each member individually assigned a weight from 1 (which meant "very low") to 5 (which meant "critical") to these two criteria for every problem. Then we computed the mean frequency and impact of every problem and rounded these to the nearest integer in the range from 1 to 5, obtaining the final weights.

The prioritization can be seen in figure 13. We addressed the problems with the highest frequency first, since this meant they were found in multiple parts of our app so they had to be resolved quickly. In the case of equal frequencies, we addressed the problems based on their impact. Else, we treated the problems equally.

4 CONCLUSIONS & IMPROVEMENTS

4.1 Conclusions

This section presents a conclusion about the main results, relating it to specific problems.

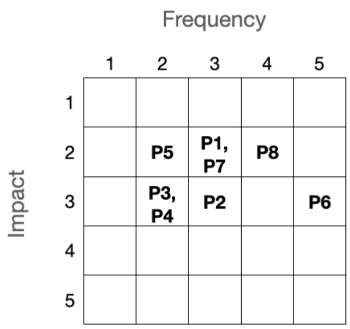

Figure 13: Severity matrix of the found problems

Our user interface proved to lack a general sense of consistency, flexibility and also directions for new users on how exactly the application works.

Firstly, about the lack of consistency, as seen in problem 6, almost all pop-ups and windows have slightly different designs that make it harder for the user to adapt to the app, therefore affecting the overall consistency of the application.

Now, in terms of flexibility, the observations in problems 1, 3 and 8 made our mistakes clear. As seen in problem 1, presetting new list titles proved to be inconvenient to users. Also, problem 2 revealed that users might want to only have to worry about board names and not their keys: right now it is difficult for new users to understand the difference between board keys and board names, and it is also harder for them to connect to boards via keys considering that if they change the board name, they are misled to believe that the key has changed as well. Problem 8 also uncovered an inflexible design for buttons, as most of them were on the right, instead of the left.

Lastly, the lack of help for new users was clarified by problems 4 and 5. As seen in problem 4, it may be confusing for new users to figure out what the label "Task List" actually refers to, when in reality it is referring to sub-tasks. This is due to internal vocabulary: during development we referred to "Tasks" as "Cards" and "Subtasks" simply as "Tasks". Problem 5 further emphasized the lack of directions, pointing out the need for a "Help" screen for users, who might feel overwhelmed at first.

4.2 Improvements

We will now present the improvements made, ordered by their priority based on figure 13. We will explain the reason behind every change, comparing it to our initial prototype and relating it to our heuristics.

We first addressed problem 6: we reviewed the entire UI, making adjustments where needed, with a heavy focus on consistency. Specifically, in figures 14 and 15 it can be seen that now all buttons, fonts and colors are consistent with the rest of the app, in conformity with heuristic 4.

To solve problem 8, we moved buttons to the left side of pop-ups, increasing reachability and overall flexibility (heuristic 7). For clear examples, see figure 16, where now the "Save" and "Cancel"

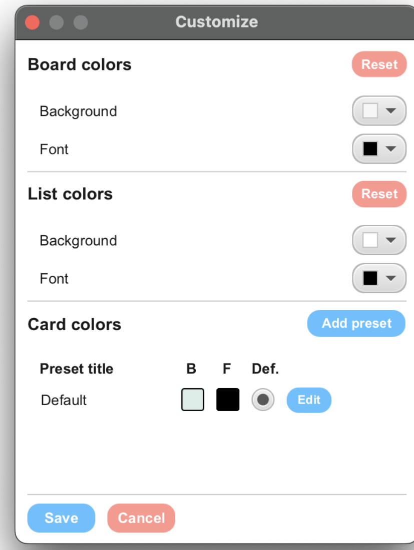

Figure 14: New customization window

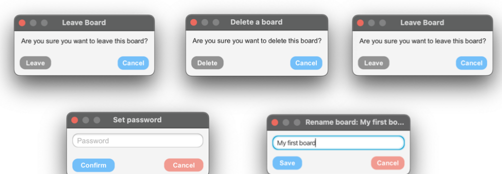

Figure 15: New pop-up designs

buttons are on the left, and also figure 15, where now the confirmation buttons are all on the left, and cancellation buttons on the right.

As pertains to problem 2, as can be seen in figure 17, a new pop-up for renaming a list was added, instead of making the title field editable: the new pop-up has a "Save" button that commits the changes and a "Cancel" button that keeps the current name. This way we also improve user control and freedom (heuristic 3).

To solve problem 7, we added icons to the left of every title in the right menu-bar of the workspace, as seen in figure 18, making everything easily recognizable by users, as required by heuristic 6.

We resolved problem 1 by popping the window in figure 17 whenever the "Add list" button is clicked, as opposed to directly adding a list with a predefined title. This way, we allowed custom titles from the beginning, increasing flexibility, as heuristic 6 requires.

Regarding the complaint in Problem 4, we redesigned the View Card and Edit/Add Card screens (figures 19, 16): now, the label reads "Subtasks" making it clear that these refers to smaller, nested tasks

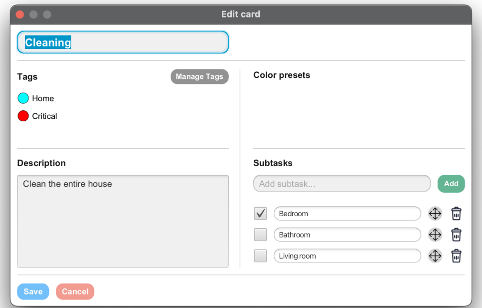

Figure 16: Updated "Edit Card" screen

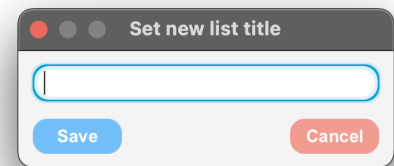

Figure 17: Improved List management

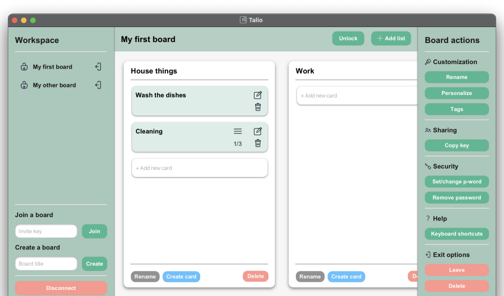

Figure 18: New main screen

to be completed in order to finish the main, original task. This change is in accordance with heuristic 10.

To resolve problem 3, we have redesigned the workspace to accommodate a new text field for the board key and a new button that allows the user to join an already existing board (figure 18). Now the user can clearly see and understand the difference between a board key and a board name, and they can also create a new board by providing only the name, while the key is automatically generated. This way, we solved the problem and improved our design according to heuristics 2 and 4.

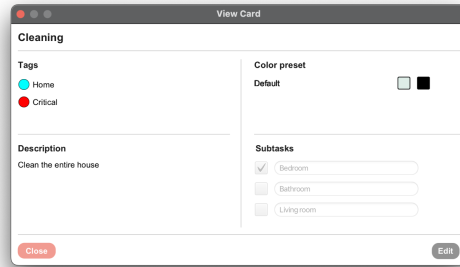

Figure 19: Updated "View Card" screen

In order to address problem 5 we implemented a new Help window that presents the user with information regarding the app (figure 20), such as the functionality of certain buttons and shortcuts. By doing this we improve yet again the level of help and documentation in our app (heuristic 10).

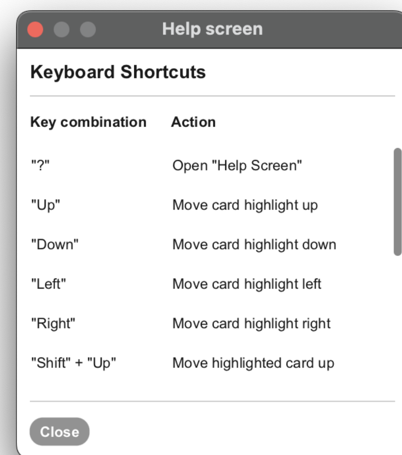

Figure 20: Help Window

Taking everything into account, the evaluation allowed us to do a wide range of improvements to the design, in conformity with our heuristics. We thank OOPP Team 29 for assessing our prototype and giving us constructive feedback.

REFERENCES

[1] Gilbert Cockton, Alan Woolrych, Lynne Hall, and Mark Hindmarch. 2004. *Changing Analysts' Tunes: The Surprising Impact of a New Instrument for Usability Inspection Method Assessment.* 145–161. https://doi.org/10.1007/978-1-4471-3754-2_9
[2] Jakob Nielsen. 1994. *Heuristic Evaluation.* John Wiley & Sons, 25–62.
[3] Jakob Nielsen. 1994. How to Conduct a Heuristic Evaluation. https://www.nngroup.com/articles/how-to-conduct-a-heuristic-evaluation. Accessed: 2023-24-03.